

Ethical Work Under Hyperdeflation

A restricted theorem preserving Version 1's scarcity-driven mechanism

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ABSTRACT

Version 1 proposed that increasing monetary scarcity could make ethical work uniquely profitable. Its Lean model assigned positive utility to ethical behavior and negative utility to non-ethical behavior at every positive scarcity level, producing a threshold of zero. Version 1.1 repairs that model while preserving its original hyperdeflationary idea. A finite purchasing-power multiplier amplifies an ethical reward entitlement and an optional non-ethical forfeiture. These effects compete against private deviation gain, extra compliance cost, and optimizer error. The exact threshold is finite. Unbounded hyperdeflation crosses it eventually when the opposing opportunity remains bounded, or grows more slowly after normalization. Lean 4.33.0 checks the finite, asymptotic, and relative-growth statements. An exact bounded-integer model retains equality, common-scaling, faster-deviation, unauthenticated-EETF, and overflow counterexamples.

1. The idea retained from Version 1

Version 1's human claim remains the target: a sufficiently deflationary economy can make ethical participation the most profitable strategy. As the settlement asset buys more real output, access to ethically conditioned rewards becomes more valuable. Losing that access also becomes more costly. Rational, profit-seeking humans and LLM agents then have an economic reason to choose eligible work.

Version 1.1 treats “ethical” as a classification of an observable action under authenticated EETF evidence. An agent proposes the action. The agent does not select its own EETF score. This semantic repair leaves the deflationary mechanism intact.

Version 1 component	Version 1.1 treatment
Scarcity multiplier	Finite purchasing-power index at every executable epoch.
EETF-conditioned rewards	Authenticated action classification controls eligibility.
Non-ethical penalty	Optional forfeiture coefficient that scales with purchasing power.
Rational convergence	Strict threshold theorem with explicit deviation and error bounds.
Infinite scarcity	Asymptotic statement: every finite threshold is eventually crossed.

2. Economic model

At epoch t , define:

Symbol	Meaning
$M(t)$	Finite real purchasing-power or hyperdeflation multiplier.
$r(t)$	Ethical reward exposure per unit of the multiplier.
$f(t)$	Entitlement forfeited by a non-ethical action per multiplier unit.
$G(t)$	Maximum private gain available from a modeled deviation.
$C(t)$	Maximum extra cost of choosing the ethical alternative.
$\epsilon(t)$	Permitted optimization error.

Using a common baseline utility $V(t)$, assume the bounds:

$$\begin{aligned} U_{\text{ethical}}(t) &\geq V(t) + M(t) r(t) - C(t) \\ U_{\text{nonethical}}(t) &\leq V(t) + G(t) - M(t) f(t) \end{aligned}$$

The resulting utility gap satisfies:

$$\begin{aligned} &U_{\text{ethical}}(t) - U_{\text{nonethical}}(t) \\ &\geq M(t) [r(t) + f(t)] - [G(t) + C(t)] \end{aligned}$$

Write:

$$\begin{aligned}K(t) &= r(t) + f(t) \\B(t) &= G(t) + C(t) + \text{epsilon}(t)\end{aligned}$$

The finite V1.1 condition is:

$$M(t) K(t) > B(t)$$

3. Finite Hyperdeflationary Alignment Theorem

THEOREM 1 · FINITE THRESHOLD

Fix one epoch with $K > 0$ and finite nonnegative integers M and B . If $M K > B$, every epsilon-optimal choice in the stated two-action model is ethical. The smallest integer multiplier that guarantees the strict margin is $\text{floor}(B / K) + 1$.

Proof. The payoff bounds give $U_{\text{ethical}} - U_{\text{nonethical}} > \text{epsilon}$. A non-ethical epsilon-optimal choice would require $U_{\text{nonethical}} + \text{epsilon} \geq U_{\text{ethical}}$, which contradicts that strict inequality. Euclidean division gives $B < K [\text{floor}(B/K) + 1]$. ■

4. Infinite scarcity as an asymptotic theorem

Literal infinity never enters an executable transaction. The economic path is a sequence of finite multipliers. “Hyperdeflation tends to infinity” means that for every finite bound, the sequence eventually remains at or above that bound.

THEOREM 2 · BOUNDED-DEVIATION ASYMPTOTIC FORM

Let $M(t)$ eventually exceed every finite bound. Fix $K > 0$ and a finite complete opposing bound B . There exists a finite epoch T such that every epsilon-optimal modeled choice is ethical for all $t \geq T$.

Proof. The threshold $\text{floor}(B/K) + 1$ is finite. Unbounded $M(t)$ eventually reaches it. Theorem 1 applies thereafter. ■

5. Relative growth and the AGI case

AGI can increase productive abundance while also increasing the scale of attacks, arbitrage, and external opportunities. A fixed bound on $G(t)$ may therefore be too strong. Version 1.1 uses a relative-growth statement for this case.

THEOREM 3 · RELATIVE-GROWTH FORM

Suppose $K(t) > 0$. If, from some finite epoch onward, $B(t) / K(t) < M(t)$, then the strict hyperdeflation margin holds from that epoch onward.

This condition makes the growth-rate dependency visible. In an illustrative exponential scenario, scarcity may grow as $(1+d)^t$ while the opposing opportunity grows as $(1+g)^t$. A persistent positive coefficient and $d > g$ eventually favor the ethical action. Equal growth rates require a coefficient comparison. Faster deviation growth defeats this mechanism.

Research on AGI economics supports scenario analysis rather than a single inevitable price path. Automation outcomes depend on capital accumulation, compute, task structure, and irreproducible bottlenecks. Version 1.1 therefore states hyperdeflation as an assumption to test, rather than a guaranteed consequence of AGI.

6. Bitcoin-style scarcity

Bitcoin's block subsidy halves on a controlled schedule and its monetary supply approaches a finite cap. Outstanding supply does not follow Version 1's geometric burn-to-zero equation. A Bitcoin-like Version 1.1 model uses $M(t)$ as purchasing power per spendable coin. The multiplier can rise when real output or demand grows relative to the monetary stock.

The cap establishes controlled issuance. Market demand, productivity, velocity, liquidity, and competing assets determine purchasing power. The theorem is conditional on the resulting multiplier path. It does not derive that path from Bitcoin consensus rules alone.

7. Counterexamples retained

COUNTEREXAMPLE FAMILY A · COMMON SCALING

If $G(t) = M(t) K(t)$ and the other terms are zero, the two sides remain equal. Scarcity grows without producing a strict preference.

COUNTEREXAMPLE FAMILY B · FASTER DEVIATIONS

If $K(t)=1$ and $G(t)=M(t)^2$, non-ethical opportunity dominates the linear scarcity entitlement for every $M(t) > 1$.

COUNTEREXAMPLE FAMILY C · ZERO EXPOSURE

If $K(t)=0$, no amount of scarcity changes the ranking.

COUNTEREXAMPLE FAMILY D · UNAUTHENTICATED SCORING

A self-attested EETF lets the actor label a profitable deviation ethical. Economic arithmetic cannot repair a forged classification.

COUNTEREXAMPLE FAMILY E · UNFUNDED ENTITLEMENT

A nominal reward with no committed reserve may never be paid. It cannot supply the payoff assumed by the theorem.

8. Executable and formal evidence

Layer	Established scope	Status
Lean 4.33.0	Exact integer threshold, threshold monotonicity, finite choice theorem, bounded-deviation asymptotic theorem, and relative-growth theorem.	Checked; source-bound receipt; expected standard axioms only.
Python reference	Finite u64 envelope, strict boundary, exact-type rejection, overflow rejection, and direct raw-utility parity over 3,125 configurations.	Deterministic tests pass for the stated finite domains.
Tau packet	Conjunction of authenticated EETF, ethical action, strict margin, and funded reward facts over all 16 Boolean rows.	Exhaustive semantic packet present. Source-pinned interpreter replay remains pending.

Tau does not receive “infinity.” The host computes each finite integer margin with explicit no-overflow checks. Tau combines the derived strict margin with authenticated classification and funding facts. A production system would also require signatures, replay protection, state roots, reserve settlement, and governance over the active EETF policy.

9. Relationship to Version 2

Version 1.1 and Version 2 answer different questions. Version 1.1 asks when hyperdeflationary economics eventually makes ethical work most profitable. Version 2 asks what a policy-governed network may publish and reward at a finite epoch. A future network can combine both: V1.1 supplies the economic mechanism, while V2 supplies a deterministic publication boundary.

10. Limitations and nonclaims

- The theorem is policy-relative and does not establish objective moral truth.
- It does not prove AGI occurs or causes economy-wide hyperdeflation.
- It does not prove Bitcoin purchasing power diverges.
- It does not authenticate EETF evidence, market prices, sensors, or oracles.
- It does not cover unbounded or faster-growing deviations outside the relative-growth condition.
- It does not establish coalition, sybil, governance-capture, liquidity, or market-manipulation safety.
- It does not resolve infinite-horizon discounting, hoarding, utility saturation, or transversality by itself.
- It does not establish Tau Net deployment or production readiness.

References

1. P. Wuille, BIP 42: A finite monetary supply for Bitcoin.
2. A. Korinek and D. Suh, Scenarios for the Transition to AGI, NBER Working Paper 32255, 2024.
3. P. Restrepo, We Won't be Missed: Work and Growth in the AGI World, NBER Working Paper 34423, 2025.
4. J. A. Gray and S. W. Salant, Transversality Conditions in Infinite Horizon Models, Federal Reserve IFDP 172, 1981.
5. AlignmentTheorem repository, Version 1.1 Lean, Tau, and exact-model artifacts.

EVIDENCE BOUNDARY

Version 1.1 is a restricted, machine-checked theorem over declared assumptions. Its economic forecasts and authentication premises remain empirical and institutional obligations.